Final Project Paper

Introduction

Our group has decided to study the public health topic of healthcare access in Ohio. We set out to do this by examining vulnerable populations and the access to healthcare options that people have available to them. We have chosen to do this by first looking at the Ohio Department of Health's data on the Ohio Health Improvement Zones. From these zones, we were able to narrow that we wanted to focus on examining the Social Vulnerability Index (SVI) that is used to determine the OHIZ. SVI data is given by county, to see which populations are at greatest risk in Ohio, with the breakdown of all different kinds of SDOH affecting this value. We have also decided to use County-level health indicators to help figure out the healthcare access vulnerabilities in the counties. Doing research on this topic is of great importance to figure out where gaps in structural support are. We have chosen to find the vulnerable populations in Ohio, and then compared the overlap between the primary care physician rates per county to determine the resource availability of healthcare access.

Data description

The first data set includes the Social Vulnerability index by county. The Social Vulnerability index per County is found by compounding many individual SDOH factors. These SDOH factors are the non-medical ways in which people work, play and live in their environment and the effect it has on health outcomes. The data set compares many factors of Socioeconomic status, including income, unemployment rate, socioeconomic score and below poverty amounts. It also has factors of built environment, including transport ability, home crowding and types of homes/availability and usage of them. The data set also includes demographic factors such as minority status, disability score, English speaking ability, and minority and senior age population.

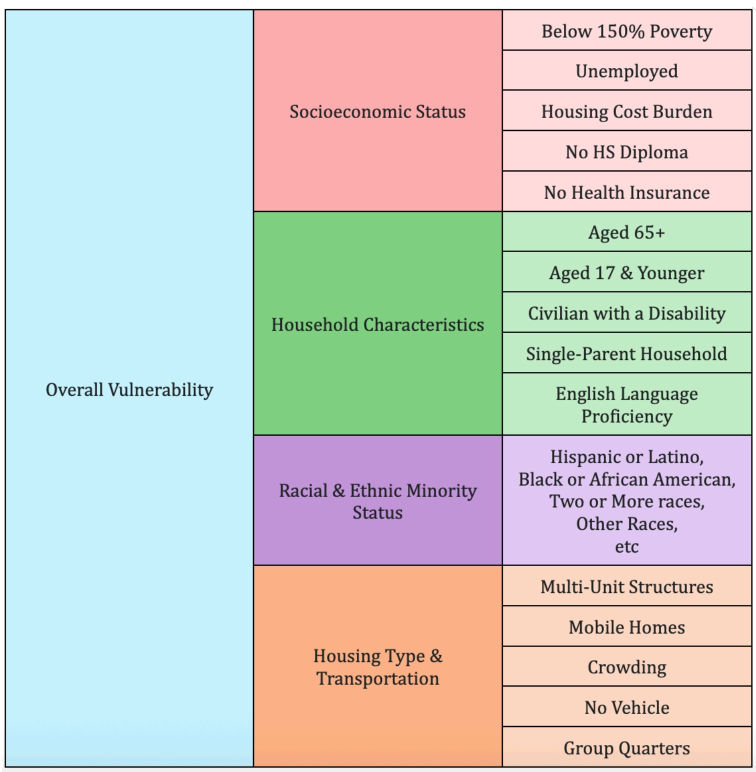
The second data set has many different county-level health indicators across all counties in Ohio. This all ranges in five different categories: Length of Life, Quality of Life, Health Infrastructure, Physical Environment ,and Social and Economic Factors. We chose to use this dataset primarily for the data it could provide on Health Care access. Even though it compares many different aspects of county-wide health, this data set provided what we could use to examine the gaps in vulnerability and support.

Project Hypothesis

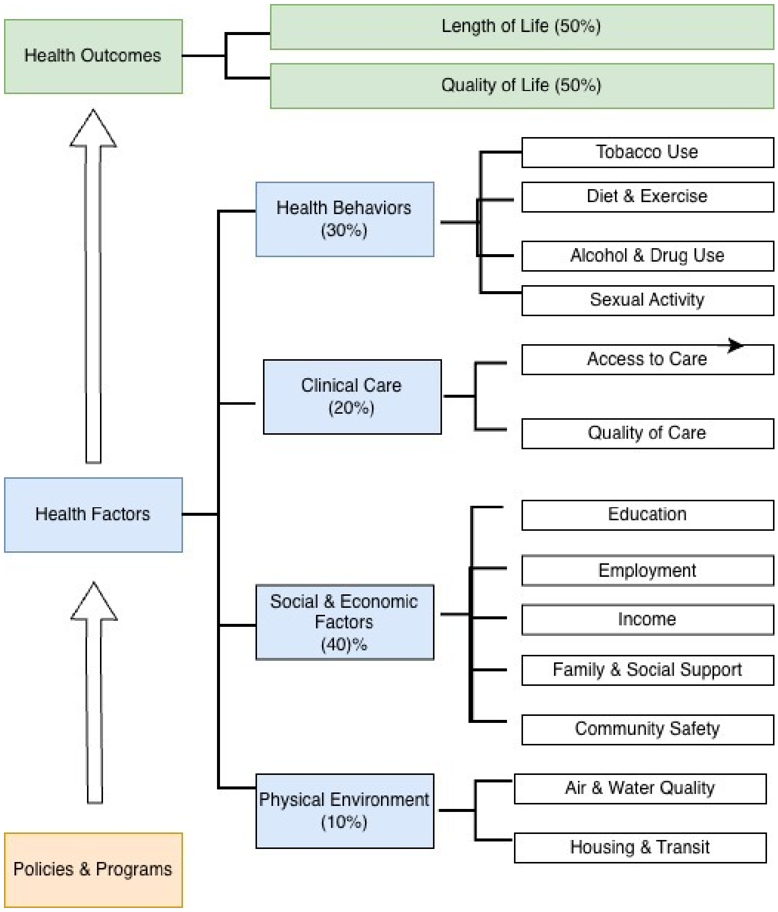
Our research question is: How does healthcare access differ across Ohio counties based on social vulnerability factors? What gaps exist between vulnerability and support? We expect that counties with higher social vulnerability will have less access to primary care physicians.

Methods

Our first dataset is from the Ohio Department of Health and is centered around Health Improvement Zones. This refers to the socioeconomic and demographic factors that can affect the resilience of individuals and the communities. The Ohio counties data that we are using has information on all 88 Counties with 16 different indicators grouped into four themes and the overall SVI. The indicators are grouped as follows:



Our second data set is from a program of the University of Wisconsin Population Health Institute which is the County Health Rankings & Roadmaps. The data is a reflection on the health of a given county, as the measures included are physical and mental well-being that are measured by the Health Outcomes of Length of Life and Quality of Life. This model is separated amongst Policies and Programs, Health Factors, and Health outcomes, and the 225 variables that fall in these for all of the Ohio Counties.



Variable Definitions:

- Dependent Variable: Primary Care Physician Rate
  - Primary Care Physician: Patients first point of contact
    - Family Practice
    - Internal Medicine
    - Pediatrics
    - OB/GYN
- Independent Variables: All variables that describe Social Vulnerability

- All our Covariates:
  * Total Population
  * Housing Units
  * Occupied Households
  * Socioeconomic Score
  * Below Poverty
  * Unemployed
  * Income
  * No high school diploma
  * House Composition & Disability Score
  * Age 65 or older
  * Age 17 or under
  * Civilian with a Disability
  * Single Parent Household
  * Minority Status Language
  * Minority
  * Speak english less than well
  * Housing type transportation
  * Multi unit structure
  * Mobile homes
  * Crowding
  * No vehicle
  * Group quarters

Exploratory Data Analysis

We will first begin with some exploratory data analysis to summarize SVI variables and health-care access measures across all the Ohio counties.

- We will first assess our dataset by creating a correlation matrix of our joined dataset to assess collinearity between our explanatory variables.
- We will then create visualizations for both data sets. We decided to create heat maps of different variables by county. These variables include some select variables we thought may be important and all four of the vulnerability indicators: Socioeconomic Status, Household Characteristics, Racial and Ethnic Minority Status, Housing type and Transportation. As well as, Primary Care Physician rate and Population to get a holistic look at the kinds of variables we would be comparing.
- We then create visuals after fitting the full model to assess its goodness of fit. These include the residual plots of the full model as well as a graph of predicted PCP rate on our full model versus the actual real PCP rate to give a better visual of our model's prediction accuracy.

Linear Regression Modeling

We fit the dataset to a linear regression model to quantify the relationship between variables that describe social vulnerability and Primary Care Physician rate.

- Full model: Yi = 0 + 1(SVI1) + 2(SVI2) +…+ 22(SVI22) + ei
  - Let Y be the Primary Care Physician Rate
  - We fit the full model with all variables that were a part of the SVI dataset. This means for example no HS diploma, aged 17 or younger, english language proficiency, and more for a total of 22 explanatory variables.

Backwards Stepwise Regression

We did backwards stepwise regression on our full model to do variable selection. We based our model criteria on AIC values.

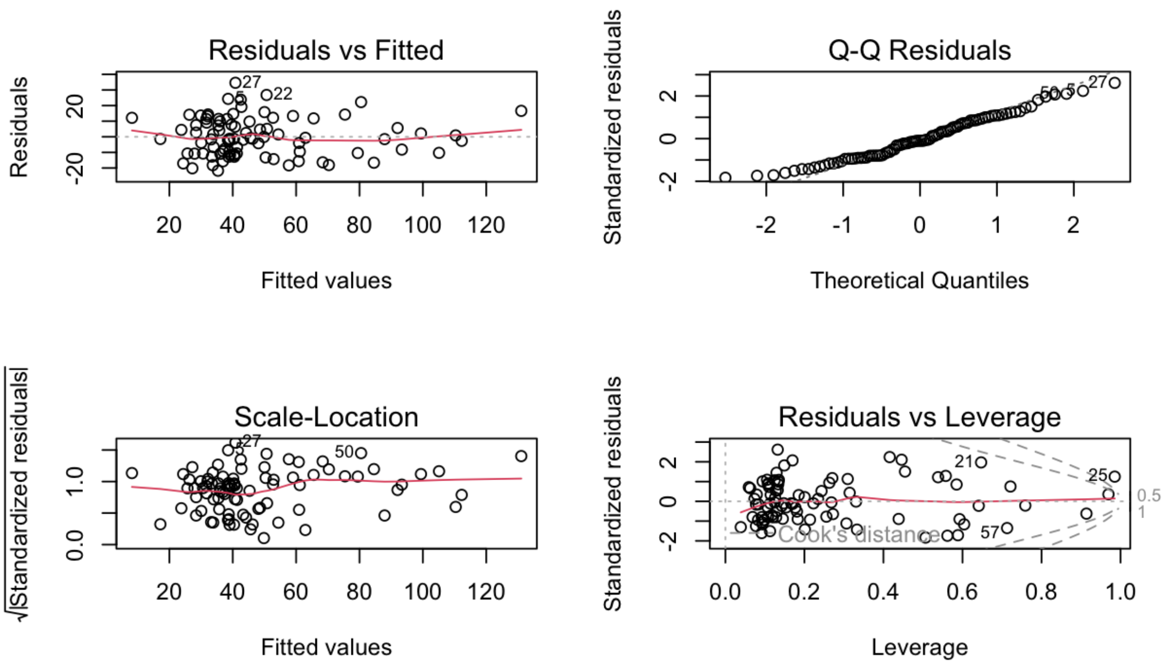
Software statement: All analyses will be conducted in R

Results

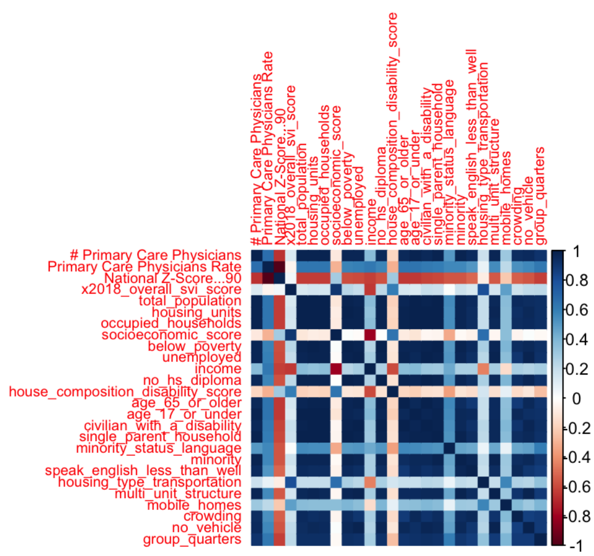
First we started with analysis of our full model.

```
Coefficients:
                                   Estimate Std. Error t value Pr(>|t|)
(Intercept)                      -5.197e+01  2.909e+01  -1.786  0.07873 .
total_population                 -2.217e-03  1.058e-03  -2.096  0.04001 *
housing_units                    -2.032e-03  1.400e-03  -1.451  0.15148
occupied_households               5.681e-03  2.332e-03   2.436  0.01760 *
socioeconomic_score              -1.588e+01  1.564e+01  -1.015  0.31374
below_poverty                     3.547e-03  1.190e-03   2.981  0.00404 **
unemployed                       -9.133e-04  4.953e-03  -0.184  0.85427
income                            2.751e-03  9.569e-04   2.875  0.00546 **
no_hs_diploma                    -4.229e-03  1.898e-03  -2.228  0.02938 *
house_composition_disability_score  1.052e+01  1.082e+01   0.973  0.33436
age_65_or_older                  -1.278e-04  1.763e-03  -0.072  0.94245
age_17_or_under                   5.141e-03  2.186e-03   2.352  0.02172 *
civilian_with_a_disability        1.517e-03  1.464e-03   1.036  0.30402
single_parent_household          -1.317e-02  6.047e-03  -2.178  0.03303 *
minority_status_language          8.571e+00  8.346e+00   1.027  0.30821
minority                          2.430e-04  7.115e-04   0.342  0.73379
speak_english_less_than_well      1.046e-03  6.749e-03   0.155  0.87731
housing_type_transportation       3.392e+01  1.186e+01   2.860  0.00570 **
multi_unit_structure             -1.365e-03  1.536e-03  -0.889  0.37731
mobile_homes                      8.254e-04  2.803e-03   0.295  0.76931
crowding                         -2.996e-02  1.153e-02  -2.598  0.01158 *
no_vehicle                       -4.401e-04  3.117e-03  -0.141  0.88817
group_quarters                    7.670e-04  1.615e-03   0.475  0.63640
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 14.2 on 65 degrees of freedom
Multiple R-squared:  0.7736,    Adjusted R-squared:  0.6969
F-statistic: 10.09 on 22 and 65 DF,  p-value: 1.533e-13
```

This model had many variables that were not significant to what we are looking for in the data. There are 22 predictors that the full model controlled for. The outcome we are looking at in this data set being the Primary Care Physician rate. It is strongly shaped by structural and demographic factors of the variables.



Next we looked at residual plots of the full model. We can see from our diagnostic plots that there are some significant trends in the graphs with clustering of points to the left and some marked outliers when there should ideally be no trends and a complete random spread. This tells us that we need to do some model assessment.



The correlation matrix shows strong clustering among many of the socioeconomic and housingrelated variables. This is particularily strong within population density and economic disadvanage. This indicates there is multicollinearity in the strucural domains, which suppoets the use of multivariable regression instead of sole predictors.

Final Model

Then we used backwards stepwise regression to perform variable selection.

```
Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)               -6.314e+01  1.743e+01  -3.622 0.000525 ***
total_population          -1.332e-03  4.877e-04  -2.731 0.007842 **
housing_units             -1.526e-03  1.110e-03  -1.375 0.173206
occupied_households        3.714e-03  1.579e-03   2.353 0.021220 *
below_poverty              3.423e-03  9.042e-04   3.786 0.000304 ***
income                     3.349e-03  6.355e-04   5.269 1.24e-06 ***
no_hs_diploma             -3.395e-03  9.823e-04  -3.456 0.000901 ***
age_17_or_under            3.523e-03  1.121e-03   3.144 0.002380 **
single_parent_household   -7.543e-03  3.629e-03  -2.078 0.041050 *
housing_type_transportation  2.864e+01  6.861e+00   4.174 7.89e-05 ***
multi_unit_structure      -1.071e-03  5.623e-04  -1.906 0.060491 .
crowding                  -2.821e-02  6.601e-03  -4.273 5.52e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13.67 on 76 degrees of freedom
Multiple R-squared:  0.7546,    Adjusted R-squared:  0.7191
F-statistic: 21.25 on 11 and 76 DF,  p-value: < 2.2e-16
```

The final model shows that poverty, educational attainment, youth population pattern, and income as the most powerful independent predictors of the primary care physician rate, even after controlling for housing conditions, household composition, and population density, explaining over 70% of the observed variance.

We are choosing to focus on 0 and 0.001 significant codes. Which leaves us with 9 variables that have the most meaningful impact on Primary Care Physician Rate.

Significant code of 0:

- Below poverty
- No High school diploma
- Age 17 or under
- Housing Type transportation

Significant code of 0.001:

- Total population
- Occupied households

- Income
- Single parent household
- Crowding

Conclusion

From looking at the final model we can see that below poverty, occupied households, income, age 17 or under, and housing type transportation all have a positive impact on the primary care physician rate. Meaning that for an increase of these variables, the expected primary care physician rate will increase. Then crowding, single parent household, total population, and no hs diploma all have a negative impact on the PCP rate. This means that as these variables increase, the expected primary care physician rate decreases. These results do support our original hypothesis.

In the field of Public Health our statistical findings show that primary care physician rates are connected to different social determinants of health (SDOH) , and they influence the overall health and well-being of a given population. SDOH are important factors in public health that affect people's daily lives and therefore their health outcomes. Our findings point to specific variables of having a higher percentage below poverty and a large population of 17 and under linking to Primary Care Physician Rates. While having a higher population without a high school diploma being linked to a decrease in the Primary Care Physician rates. This means in the context of public health there are factors that professionals in this field can work to fix and see whether the outcome changes in these areas.

Our limitations include that the two datasets are from different years, the SVI data is from 2021 and the Ohio County data is from 2025. This might not be a full representation of the current variables that are affecting the primary care physician rate. The Ohio county data does not define what primary care physicians include. So in this case, we are assuming that this does not count the households that may use emergency rooms as their acute care instead of going to a primary care practice. There also might be other factors that may be impacting the primary care physician rate. The cross sectionality of the data makes it difficult to determine if the variables that we have chosen are the ones that are actually causing the effects. Not only could there be limitations within the dataset, but we are also only looking at county-level data. There could be inequality within the counties themselves.

Some things to do in the future to help increase the Primary Care Physician Rate includes strengthening education and health literacy programs in counties with a low high school diploma rate. This is because as we can see from the results, having a high number of people with no high school diploma has a negative effect on the primary care physician rate. Another way to help increase access through the PCP rate is by developing support services for single-parent households. By giving more support to these households, it can help decrease the number of people that are struggling with access.

Appendix

Here is an appendix of all our code.

```
# Imports
library(ggplot2)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v forcats   1.0.1     v stringr   1.6.0
v lubridate 1.9.4     v tibble    3.3.0
v purrr     1.2.0     v tidyr     1.3.1
v readr     2.1.5
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
library(readxl)
library(readr)
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:purrr':

    some

The following object is masked from 'package:dplyr':

    recode
```

```r
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```r
library(ggrepel)
```

Setting up dataset

Import SVI dataset

```r
# SVI Dataset Import
county_raw = read_csv("../data/SVI_County_Export.csv", show_col_types = FALSE) |>
  clean_names()

# SVI dataset cleaning
county_svi = county_raw |>
  separate(location, into = c("county_name", "state"), sep = ", ") |>
  mutate(
    county = county_name |>
      str_remove(" County") |>
      str_to_upper()
  )
prep = function(df) {
  df |>
    mutate(across(where(is.character), as.factor)) |>
    mutate(across(where(is.factor), ~fct_explicit_na(.x, na_level = "Missing"))) |>
    mutate(across(where(is.numeric), ~ifelse(is.na(.x), median(.x, na.rm = TRUE), .x)))
}

county_svi_simple = prep(county_svi)
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `across(where(is.factor), ~fct_explicit_na(.x, na_level =
  "Missing"))`.
Caused by warning:
! `fct_explicit_na()` was deprecated in forcats 1.0.0.
i Please use `fct_na_value_to_level()` instead.
```

```r
county_svi_simple$county_name <- gsub(" County$", "", county_svi_simple$county_name)
county_svi_simple$fips <- sprintf("%05d", county_svi_simple$fips)
head(county_svi_simple)
```

```
# A tibble: 6 x 27
  county_name state fips  x2018_overall_svi_score total_population housing_units
  <chr>       <fct> <chr>                   <dbl>            <dbl>         <dbl>
1 Adams       Ohio  39001                   0.931            27878         12920
2 Allen       Ohio  39003                   0.770           103642         45063
3 Ashland     Ohio  39005                   0.575            53477         22250
4 Ashtabula   Ohio  39007                   1                98136         46107
5 Athens      Ohio  39009                   0.942            65936         26610
6 Auglaize    Ohio  39011                   0.023            45784         19852
```

```
# i 21 more variables: occupied_households <dbl>, socioeconomic_score <dbl>,
#   below_poverty <dbl>, unemployed <dbl>, income <dbl>, no_hs_diploma <dbl>,
#   house_composition_disability_score <dbl>, age_65_or_older <dbl>,
#   age_17_or_under <dbl>, civilian_with_a_disability <dbl>,
#   single_parent_household <dbl>, minority_status_language <dbl>,
#   minority <dbl>, speak_english_less_than_well <dbl>,
#   housing_type_transportation <dbl>, multi_unit_structure <dbl>, ...
```

Import Healthcare Access Dataset

```
# Healthcare access dataset
excel_sheets("../data/2025 County Health Rankings Ohio Data - v3.xlsx")
```

```
[1] "Introduction"                    "Select Measure Data"
[3] "Select Measure Sources & Years" "Additional Measure Data"
[5] "Addtl Measure Sources & Years"  "Health Groups"
```

```
main <- suppressMessages(read_excel("../data/2025 County Health Rankings Ohio Data - v3.xlsx'
                       sheet = "Select Measure Data", skip = 1))
healthcare_vars <- c(
  "FIPS", "State", "County",  # keep these for identification

  "# Primary Care Physicians",
  "Primary Care Physicians Rate",
  "Primary Care Physicians Ratio",
  "National Z-Score...90"
)

healthcare_access_data <- main[, healthcare_vars]
healthcare_access_data <- healthcare_access_data[-1, ]
```

Join datasets

```
#inner join the two datasets by county, wsanity checked 54 colns
merged_data = healthcare_access_data %>%
  left_join(county_svi_simple, by = c("FIPS" = "fips"))

head(merged_data)
```

```
# A tibble: 6 x 33
```

```
  FIPS  State County    `# Primary Care Physicians` Primary Care Physicians Ra~1
  <chr> <chr> <chr>                           <dbl>                        <dbl>
1 39001 Ohio  Adams                              12                         43.6
2 39003 Ohio  Allen                              66                         64.9
3 39005 Ohio  Ashland                            23                         44.0
4 39007 Ohio  Ashtabula                          28                         28.8
5 39009 Ohio  Athens                             39                         62.8
6 39011 Ohio  Auglaize                           16                         34.7
# i abbreviated name: 1: `Primary Care Physicians Rate`
# i 28 more variables: `Primary Care Physicians Ratio` <chr>,
#   `National Z-Score...90` <dbl>, county_name <chr>, state <fct>,
#   x2018_overall_svi_score <dbl>, total_population <dbl>, housing_units <dbl>,
#   occupied_households <dbl>, socioeconomic_score <dbl>, below_poverty <dbl>,
#   unemployed <dbl>, income <dbl>, no_hs_diploma <dbl>,
#   house_composition_disability_score <dbl>, age_65_or_older <dbl>, ...
```

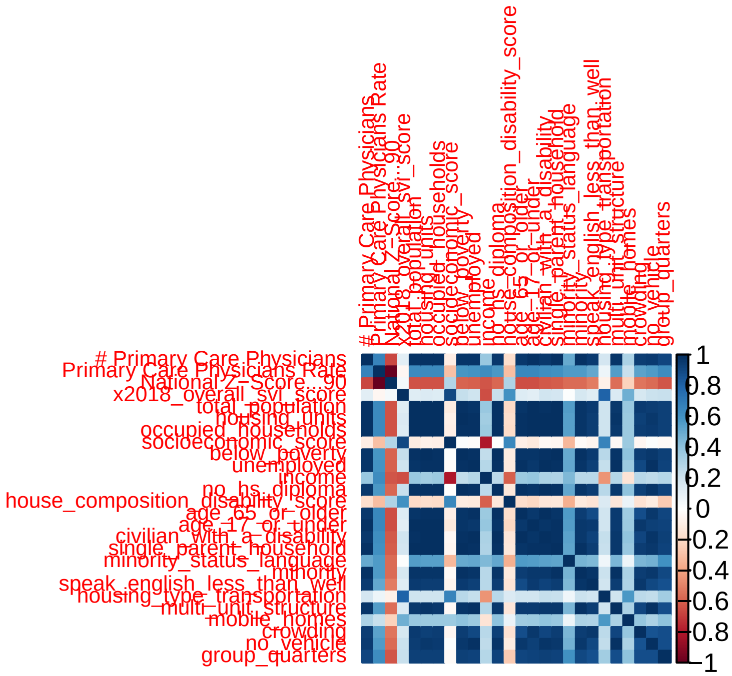
```r
print(colnames(merged_data))
```

```
 [1] "FIPS"                               "State"
 [3] "County"                             "# Primary Care Physicians"
 [5] "Primary Care Physicians Rate"       "Primary Care Physicians Ratio"
 [7] "National Z-Score...90"              "county_name"
 [9] "state"                              "x2018_overall_svi_score"
[11] "total_population"                   "housing_units"
[13] "occupied_households"                "socioeconomic_score"
[15] "below_poverty"                      "unemployed"
[17] "income"                             "no_hs_diploma"
[19] "house_composition_disability_score" "age_65_or_older"
[21] "age_17_or_under"                    "civilian_with_a_disability"
[23] "single_parent_household"            "minority_status_language"
[25] "minority"                           "speak_english_less_than_well"
[27] "housing_type_transportation"        "multi_unit_structure"
[29] "mobile_homes"                       "crowding"
[31] "no_vehicle"                         "group_quarters"
[33] "county"
```

EDA

Correlation matrix

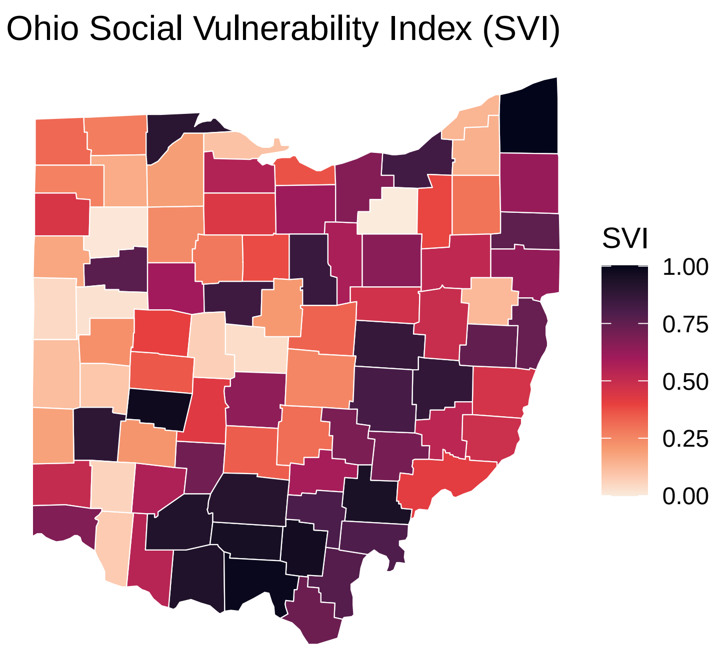
```
numeric_vars <- dplyr::select(merged_data, where(is.numeric))
cor_matrix = cor(numeric_vars, use = "pairwise.complete.obs")
corrplot(cor_matrix, method = "color", tl.cex = 0.7)
```



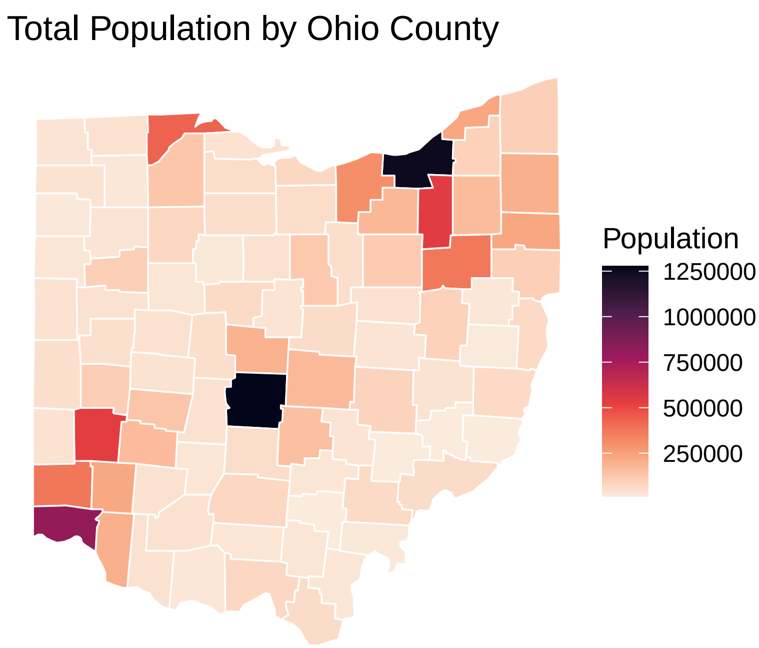
Heat Maps

```
ohio_map <- map_data("county") %>%
  filter(region == "ohio") %>%
  mutate(county = toupper(subregion))

map_df <- ohio_map %>%
  left_join(merged_data, by = "county")

# Heat map: Ohio SVI score
ggplot(map_df, aes(long, lat, group = group, fill = x2018_overall_svi_score)) +
  geom_polygon(color = "white", size = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket",  direction = -1) +
  theme_void() +
  labs(fill = "SVI", title = "Ohio Social Vulnerability Index (SVI)")
```

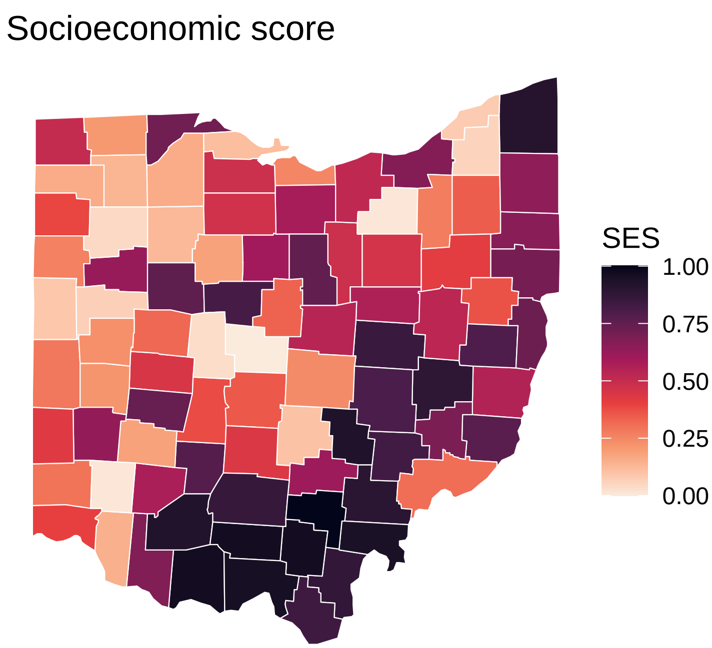
```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```



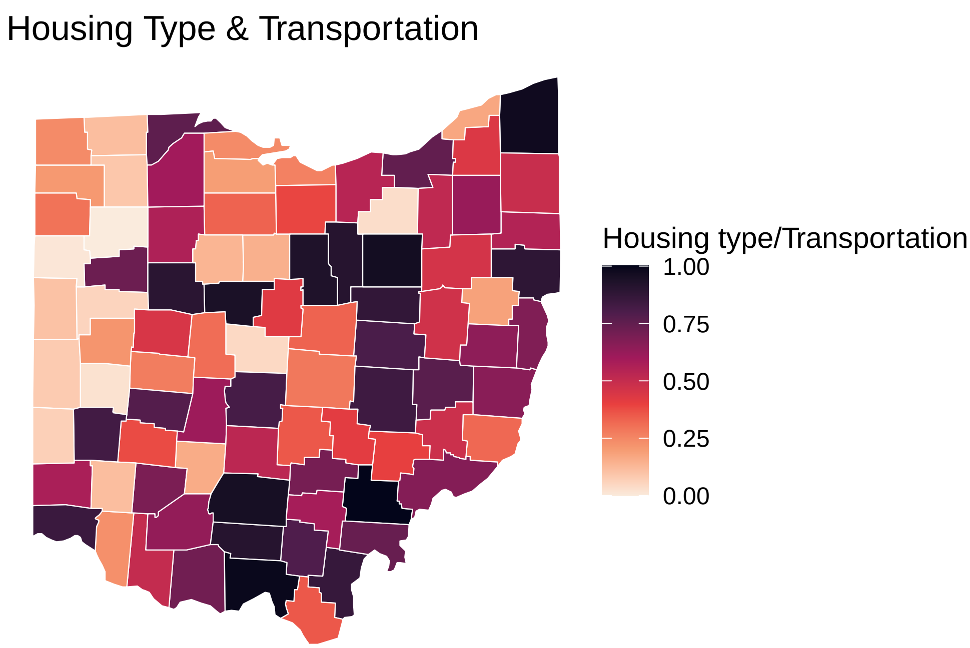

```r
# Heat map: Total population
ggplot(map_df, aes(long, lat, group = group, fill = total_population)) +
  geom_polygon(color = "white", linewidth = 0.3) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(
    title = "Total Population by Ohio County",
    fill = "Population"
  )
```




```
# Heat map: Socioeconomic Score
ggplot(map_df, aes(long, lat, group = group, fill = socioeconomic_score)) +
  geom_polygon(color = "white", size = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket",  direction = -1) +
  theme_void() +
  labs(fill = "SES", title = "Socioeconomic score")
```

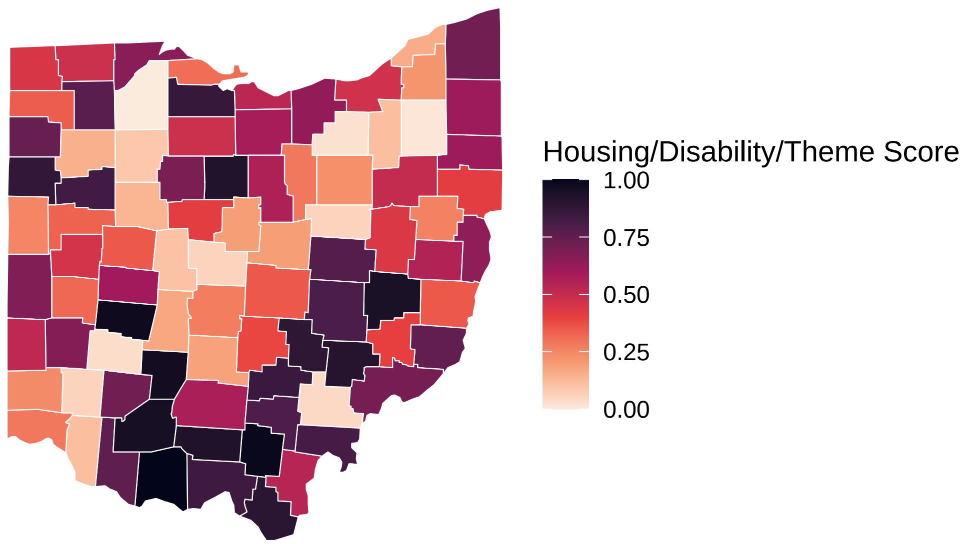



```
# Heat map: Housing Type & Transportation
ggplot(map_df, aes(long, lat, group = group, fill = housing_type_transportation)) +
  geom_polygon(color = "white", size = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket",  direction = -1) +
  theme_void() +
  labs(fill = "Housing type/Transportation", title = "Housing Type & Transportation")
```

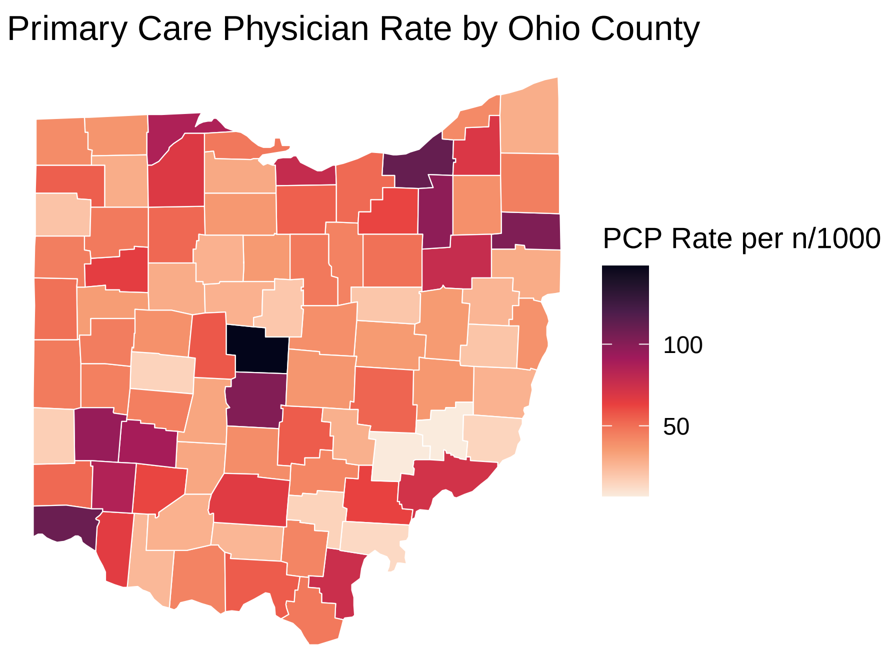



```r
# Heat map: Household Composition & Disability Vulnerability
ggplot(map_df, aes(long, lat, group = group, fill = house_composition_disability_score)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(
    title =  "Household Composition & Disability Vulnerability\nby Ohio County",
    fill = "Housing/Disability/Theme Score",
  )
```

Household Composition & Disability Vulnerability
by Ohio County




```
# Heat map: Primary Care Physician Rate
ggplot(map_df, aes(long, lat, group = group, fill = `Primary Care Physicians Rate`)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(
    title =  "Primary Care Physician Rate by Ohio County",
    fill = "PCP Rate per n/1000",
  )
```



Linear Model

```
full_mod = lm(
  `Primary Care Physicians Rate` ~
    total_population +
    housing_units +
    occupied_households +
    socioeconomic_score +
    below_poverty +
    unemployed +
    income +
    no_hs_diploma +
    house_composition_disability_score +
    age_65_or_older +
    age_17_or_under +
    civilian_with_a_disability +
    single_parent_household +
    minority_status_language +
    minority +
    speak_english_less_than_well +
    housing_type_transportation+
    multi_unit_structure +
```

```r
    mobile_homes +
    crowding +
    no_vehicle +
    group_quarters,
  data = merged_data
)
summary(full_mod)
```

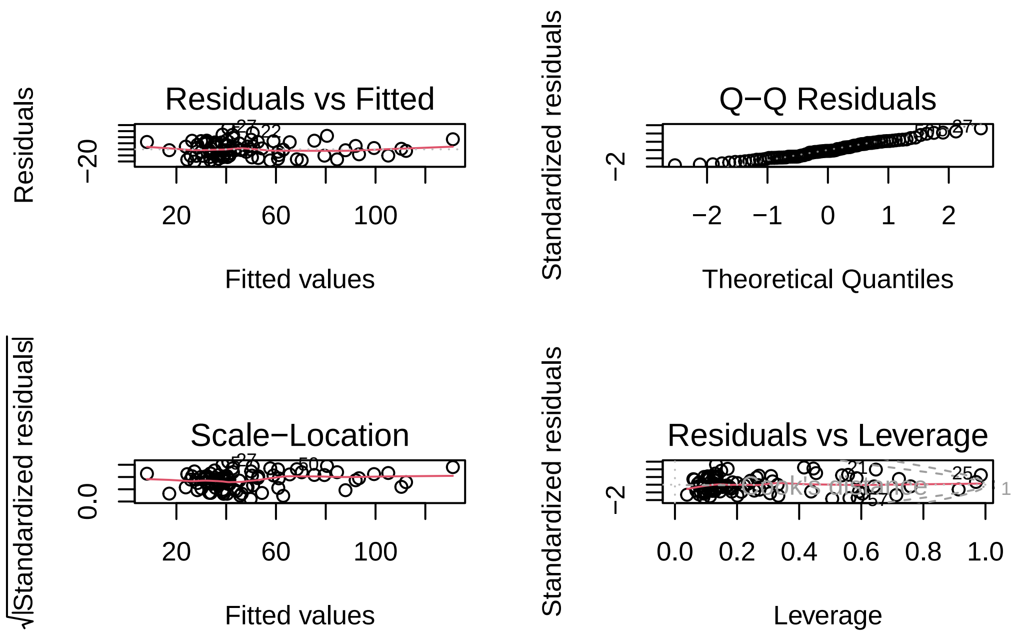
```
Call:
lm(formula = `Primary Care Physicians Rate` ~ total_population +
    housing_units + occupied_households + socioeconomic_score +
    below_poverty + unemployed + income + no_hs_diploma + house_composition_disability_score
    age_65_or_older + age_17_or_under + civilian_with_a_disability +
    single_parent_household + minority_status_language + minority +
    speak_english_less_than_well + housing_type_transportation +
    multi_unit_structure + mobile_homes + crowding + no_vehicle +
    group_quarters, data = merged_data)

Residuals:
    Min      1Q  Median      3Q     Max
-21.729 -10.641  -1.293   9.791  34.610

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)
(Intercept)                       -5.197e+01  2.909e+01  -1.786  0.07873 .
total_population                  -2.217e-03  1.058e-03  -2.096  0.04001 *
housing_units                     -2.032e-03  1.400e-03  -1.451  0.15148
occupied_households                5.681e-03  2.332e-03   2.436  0.01760 *
socioeconomic_score               -1.588e+01  1.564e+01  -1.015  0.31374
below_poverty                      3.547e-03  1.190e-03   2.981  0.00404 **
unemployed                        -9.133e-04  4.953e-03  -0.184  0.85427
income                             2.751e-03  9.569e-04   2.875  0.00546 **
no_hs_diploma                     -4.229e-03  1.898e-03  -2.228  0.02938 *
house_composition_disability_score  1.052e+01  1.082e+01   0.973  0.33436
age_65_or_older                   -1.278e-04  1.763e-03  -0.072  0.94245
age_17_or_under                    5.141e-03  2.186e-03   2.352  0.02172 *
civilian_with_a_disability         1.517e-03  1.464e-03   1.036  0.30402
single_parent_household           -1.317e-02  6.047e-03  -2.178  0.03303 *
minority_status_language           8.571e+00  8.346e+00   1.027  0.30821
minority                           2.430e-04  7.115e-04   0.342  0.73379
speak_english_less_than_well       1.046e-03  6.749e-03   0.155  0.87731
```

```
housing_type_transportation         3.392e+01  1.186e+01   2.860  0.00570 **
multi_unit_structure               -1.365e-03  1.536e-03  -0.889  0.37731
mobile_homes                        8.254e-04  2.803e-03   0.295  0.76931
crowding                           -2.996e-02  1.153e-02  -2.598  0.01158 *
no_vehicle                         -4.401e-04  3.117e-03  -0.141  0.88817
group_quarters                      7.670e-04  1.615e-03   0.475  0.63640
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.2 on 65 degrees of freedom
Multiple R-squared:  0.7736,    Adjusted R-squared:  0.6969
F-statistic: 10.09 on 22 and 65 DF,  p-value: 1.533e-13
```

Full Model Visualizations

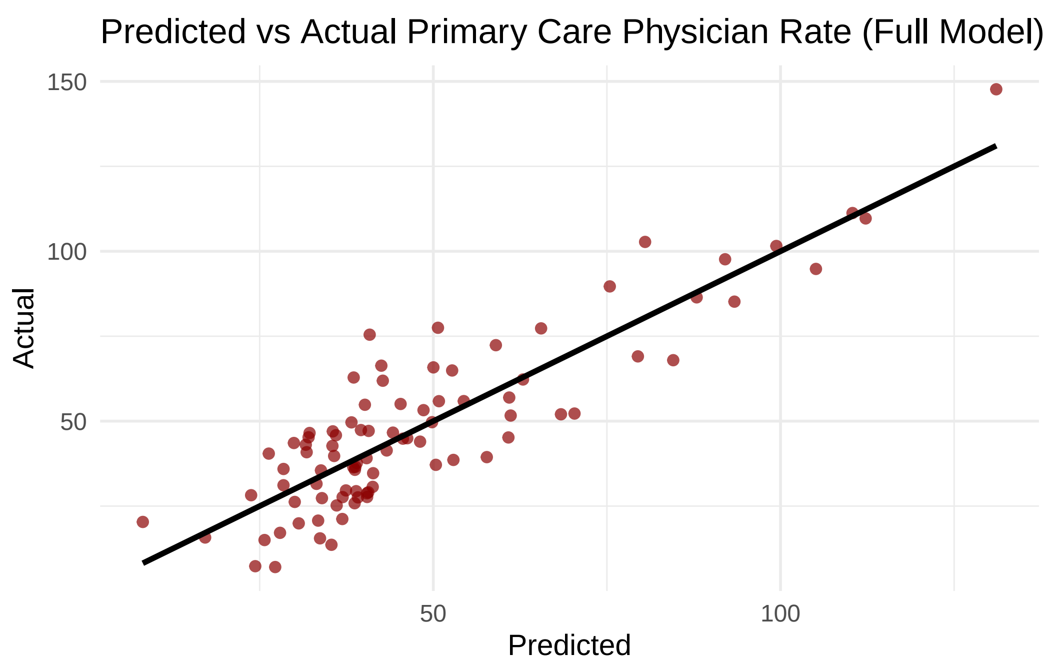
```r
# Residuals plots
par(mfrow = c(2,2))
plot(full_mod)
```

```
Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
```



```r
# PCP Prediction plot
merged_data$pred_full <- predict(full_mod)

ggplot(merged_data,
       aes(x = pred_full, y = `Primary Care Physicians Rate`)) +
  geom_point(alpha = 0.7, color = "darkred") +
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  labs(
    title = "Predicted vs Actual Primary Care Physician Rate (Full Model)",
    x = "Predicted",
    y = "Actual"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Backwards Stepwise Regression

```r
step_mod = stepAIC(full_mod, direction = "backward")
```

```
Start:  AIC=486.36
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + socioeconomic_score + below_poverty +
    unemployed + income + no_hs_diploma + house_composition_disability_score +
    age_65_or_older + age_17_or_under + civilian_with_a_disability +
    single_parent_household + minority_status_language + minority +
    speak_english_less_than_well + housing_type_transportation +
    multi_unit_structure + mobile_homes + crowding + no_vehicle +
    group_quarters

                                     Df Sum of Sq   RSS    AIC
- age_65_or_older                     1      1.06 13115 484.37
- no_vehicle                          1      4.02 13118 484.39
- speak_english_less_than_well        1      4.85 13119 484.39
- unemployed                          1      6.86 13121 484.41
- mobile_homes                        1     17.50 13131 484.48
- minority                            1     23.54 13137 484.52
- group_quarters                      1     45.51 13159 484.66
- multi_unit_structure                1    159.43 13273 485.42
- house_composition_disability_score  1    190.85 13305 485.63
- socioeconomic_score                 1    207.96 13322 485.74
- minority_status_language            1    212.81 13327 485.78
- civilian_with_a_disability          1    216.56 13330 485.80
<none>                                            13114 486.36
- housing_units                       1    425.01 13539 487.17
- total_population                    1    886.03 14000 490.11
- single_parent_household             1    957.17 14071 490.56
- no_hs_diploma                       1   1001.10 14115 490.83
- age_17_or_under                     1   1116.01 14230 491.55
- occupied_households                 1   1197.28 14311 492.05
- crowding                            1   1362.01 14476 493.06
- housing_type_transportation         1   1649.74 14764 494.79
- income                              1   1667.58 14782 494.89
- below_poverty                       1   1793.16 14907 495.64

Step:  AIC=484.37
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + socioeconomic_score + below_poverty +
    unemployed + income + no_hs_diploma + house_composition_disability_score +
    age_17_or_under + civilian_with_a_disability + single_parent_household +
    minority_status_language + minority + speak_english_less_than_well +
    housing_type_transportation + multi_unit_structure + mobile_homes +
    crowding + no_vehicle + group_quarters
```

```
                                   Df Sum of Sq   RSS    AIC
- no_vehicle                        1      5.14 13120 482.40
- speak_english_less_than_well      1      5.64 13121 482.40
- unemployed                        1      7.94 13123 482.42
- mobile_homes                      1     26.32 13141 482.54
- minority                          1     27.26 13142 482.55
- group_quarters                    1     46.26 13161 482.68
- multi_unit_structure              1    167.95 13283 483.49
- house_composition_disability_score  1    196.28 13311 483.67
- socioeconomic_score               1    207.74 13323 483.75
- minority_status_language          1    211.75 13327 483.78
- civilian_with_a_disability        1    228.68 13344 483.89
<none>                                          13115 484.37
- housing_units                     1    498.20 13613 485.65
- total_population                  1    886.34 14001 488.12
- single_parent_household           1   1052.40 14167 489.16
- age_17_or_under                   1   1194.86 14310 490.04
- occupied_households               1   1243.88 14359 490.34
- no_hs_diploma                     1   1250.09 14365 490.38
- crowding                          1   1411.46 14526 491.36
- housing_type_transportation       1   1648.71 14764 492.79
- income                            1   1724.32 14839 493.24
- below_poverty                     1   1815.14 14930 493.77

Step:  AIC=482.4
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + socioeconomic_score + below_poverty +
    unemployed + income + no_hs_diploma + house_composition_disability_score +
    age_17_or_under + civilian_with_a_disability + single_parent_household +
    minority_status_language + minority + speak_english_less_than_well +
    housing_type_transportation + multi_unit_structure + mobile_homes +
    crowding + group_quarters

                                   Df Sum of Sq   RSS    AIC
- unemployed                        1     13.26 13133 480.49
- speak_english_less_than_well      1     16.28 13136 480.51
- minority                          1     22.51 13143 480.55
- mobile_homes                      1     38.13 13158 480.66
- group_quarters                    1     48.85 13169 480.73
- house_composition_disability_score  1    194.37 13314 481.70
- multi_unit_structure              1    195.77 13316 481.70
- socioeconomic_score               1    202.60 13323 481.75
```

```
- minority_status_language           1    207.62 13328 481.78
- civilian_with_a_disability         1    236.57 13357 481.97
<none>                                           13120 482.40
- housing_units                      1    517.57 13638 483.81
- total_population                   1    971.99 14092 486.69
- age_17_or_under                    1   1207.54 14328 488.15
- occupied_households                1   1351.19 14471 489.03
- single_parent_household            1   1355.59 14476 489.05
- crowding                           1   1438.36 14558 489.56
- housing_type_transportation        1   1754.31 14874 491.45
- no_hs_diploma                      1   1785.87 14906 491.63
- income                             1   1786.68 14907 491.64
- below_poverty                      1   1828.20 14948 491.88

Step:  AIC=480.49
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + socioeconomic_score + below_poverty +
    income + no_hs_diploma + house_composition_disability_score +
    age_17_or_under + civilian_with_a_disability + single_parent_household +
    minority_status_language + minority + speak_english_less_than_well +
    housing_type_transportation + multi_unit_structure + mobile_homes +
    crowding + group_quarters

                                    Df Sum of Sq   RSS    AIC
- speak_english_less_than_well       1     16.27 13150 478.60
- minority                           1     19.22 13153 478.62
- group_quarters                     1     50.29 13184 478.83
- mobile_homes                       1     53.39 13187 478.85
- house_composition_disability_score  1    189.64 13323 479.75
- socioeconomic_score                1    198.82 13332 479.81
- minority_status_language           1    215.65 13349 479.92
- civilian_with_a_disability         1    226.53 13360 480.00
- multi_unit_structure               1    237.28 13371 480.07
<none>                                           13133 480.49
- housing_units                      1    561.30 13695 482.17
- total_population                   1   1041.75 14175 485.21
- age_17_or_under                    1   1279.36 14413 486.67
- single_parent_household            1   1376.95 14510 487.26
- occupied_households                1   1471.66 14605 487.84
- crowding                           1   1671.57 14805 489.03
- housing_type_transportation        1   1752.21 14886 489.51
- income                             1   1778.39 14912 489.67
- no_hs_diploma                      1   1832.88 14966 489.99
```

```
- below_poverty                       1   1878.57 15012 490.26

Step:  AIC=478.6
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + socioeconomic_score + below_poverty +
    income + no_hs_diploma + house_composition_disability_score +
    age_17_or_under + civilian_with_a_disability + single_parent_household +
    minority_status_language + minority + housing_type_transportation +
    multi_unit_structure + mobile_homes + crowding + group_quarters

                                     Df Sum of Sq   RSS    AIC
- group_quarters                      1     34.05 13184 476.83
- mobile_homes                        1     63.96 13214 477.03
- house_composition_disability_score  1    200.88 13350 477.93
- civilian_with_a_disability          1    210.51 13360 478.00
- socioeconomic_score                 1    224.39 13374 478.09
- minority_status_language            1    238.39 13388 478.18
- minority                            1    264.02 13414 478.35
<none>                                            13150 478.60
- multi_unit_structure                1    609.66 13759 480.59
- housing_units                       1    617.99 13768 480.64
- occupied_households                 1   1463.65 14613 485.89
- single_parent_household             1   1579.51 14729 486.58
- total_population                    1   1646.51 14796 486.98
- housing_type_transportation         1   1744.38 14894 487.56
- crowding                            1   1828.80 14978 488.06
- income                              1   1830.17 14980 488.07
- below_poverty                       1   1899.03 15049 488.47
- no_hs_diploma                       1   2004.04 15154 489.08
- age_17_or_under                     1   2211.05 15361 490.28

Step:  AIC=476.83
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + socioeconomic_score + below_poverty +
    income + no_hs_diploma + house_composition_disability_score +
    age_17_or_under + civilian_with_a_disability + single_parent_household +
    minority_status_language + minority + housing_type_transportation +
    multi_unit_structure + mobile_homes + crowding

                                     Df Sum of Sq   RSS    AIC
- mobile_homes                        1     47.34 13231 475.14
- house_composition_disability_score  1    168.10 13352 475.94
- socioeconomic_score                 1    216.57 13400 476.26
```

```
- civilian_with_a_disability          1    241.93 13426 476.43
- minority                            1    268.26 13452 476.60
- minority_status_language            1    276.68 13460 476.65
<none>                                            13184 476.83
- housing_units                       1    599.79 13784 478.74
- multi_unit_structure                1    621.04 13805 478.88
- occupied_households                 1   1465.55 14649 484.10
- single_parent_household             1   1545.79 14730 484.58
- total_population                    1   1739.23 14923 485.73
- income                              1   1821.85 15006 486.22
- crowding                            1   1866.16 15050 486.48
- below_poverty                       1   1978.11 15162 487.13
- housing_type_transportation         1   2222.76 15406 488.54
- age_17_or_under                     1   2226.88 15410 488.56
- no_hs_diploma                       1   2338.85 15522 489.20

Step:  AIC=475.14
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + socioeconomic_score + below_poverty +
    income + no_hs_diploma + house_composition_disability_score +
    age_17_or_under + civilian_with_a_disability + single_parent_household +
    minority_status_language + minority + housing_type_transportation +
    multi_unit_structure + crowding

                                     Df Sum of Sq   RSS    AIC
- house_composition_disability_score  1    216.86 13448 474.57
- socioeconomic_score                 1    218.86 13450 474.59
- minority                            1    231.02 13462 474.67
- minority_status_language            1    234.99 13466 474.69
- civilian_with_a_disability          1    250.84 13482 474.80
<none>                                            13231 475.14
- housing_units                       1    562.07 13793 476.80
- multi_unit_structure                1    575.18 13806 476.89
- occupied_households                 1   1438.03 14669 482.22
- single_parent_household             1   1623.08 14854 483.33
- total_population                    1   1700.20 14931 483.78
- crowding                            1   1819.74 15051 484.48
- income                              1   1959.24 15190 485.29
- age_17_or_under                     1   2199.96 15431 486.68
- no_hs_diploma                       1   2381.09 15612 487.70
- housing_type_transportation         1   2522.31 15753 488.50
- below_poverty                       1   2649.10 15880 489.20
```

```
Step:  AIC=474.57
`Primary Care Physicians Rate` ~ total_population + housing_units + 
    occupied_households + socioeconomic_score + below_poverty + 
    income + no_hs_diploma + age_17_or_under + civilian_with_a_disability + 
    single_parent_household + minority_status_language + minority + 
    housing_type_transportation + multi_unit_structure + crowding

                              Df Sum of Sq   RSS    AIC
- socioeconomic_score          1     66.27 13514 473.01
- minority_status_language     1    239.16 13687 474.12
- minority                     1    287.07 13735 474.43
<none>                                     13448 474.57
- civilian_with_a_disability   1    431.02 13879 475.35
- housing_units                1    589.13 14037 476.35
- multi_unit_structure         1    689.81 14138 476.98
- single_parent_household      1   1476.53 14924 481.74
- occupied_households          1   1566.88 15015 482.27
- crowding                     1   1629.72 15078 482.64
- income                       1   1941.06 15389 484.44
- total_population             1   2077.53 15525 485.21
- housing_type_transportation  1   2348.57 15796 486.74
- age_17_or_under              1   2387.63 15836 486.96
- below_poverty                1   2446.83 15895 487.28
- no_hs_diploma                1   2543.88 15992 487.82

Step:  AIC=473.01
`Primary Care Physicians Rate` ~ total_population + housing_units + 
    occupied_households + below_poverty + income + no_hs_diploma + 
    age_17_or_under + civilian_with_a_disability + single_parent_household + 
    minority_status_language + minority + housing_type_transportation + 
    multi_unit_structure + crowding

                              Df Sum of Sq   RSS    AIC
- minority                     1     243.5 13758 472.58
<none>                                     13514 473.01
- minority_status_language     1     330.1 13844 473.13
- civilian_with_a_disability   1     389.4 13904 473.51
- housing_units                1     553.6 14068 474.54
- multi_unit_structure         1     635.2 14149 475.05
- single_parent_household      1    1411.4 14926 479.75
- occupied_households          1    1502.4 15016 480.28
- crowding                     1    1597.6 15112 480.84
- total_population             1    2028.4 15542 483.31
```

```
- below_poverty               1    2390.5 15905 485.34
- age_17_or_under             1    2417.7 15932 485.49
- no_hs_diploma               1    2479.1 15993 485.83
- housing_type_transportation  1    2858.6 16373 487.89
- income                      1    3920.5 17435 493.42

Step:  AIC=472.58
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + below_poverty + income + no_hs_diploma +
    age_17_or_under + civilian_with_a_disability + single_parent_household +
    minority_status_language + housing_type_transportation +
    multi_unit_structure + crowding

                              Df Sum of Sq   RSS    AIC
- civilian_with_a_disability   1     249.2 14007 472.16
- minority_status_language     1     292.4 14050 472.43
<none>                                     13758 472.58
- housing_units               1     419.8 14177 473.22
- multi_unit_structure        1     420.6 14178 473.23
- single_parent_household     1    1179.2 14937 477.81
- occupied_households         1    1262.8 15020 478.31
- crowding                    1    1465.0 15223 479.48
- total_population            1    1797.2 15555 481.38
- age_17_or_under             1    2242.0 16000 483.86
- no_hs_diploma               1    2313.0 16071 484.25
- housing_type_transportation  1    2622.6 16380 485.93
- below_poverty               1    2735.0 16493 486.53
- income                      1    4121.3 17879 493.64

Step:  AIC=472.16
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + below_poverty + income + no_hs_diploma +
    age_17_or_under + single_parent_household + minority_status_language +
    housing_type_transportation + multi_unit_structure + crowding

                              Df Sum of Sq   RSS    AIC
- minority_status_language     1     205.4 14212 471.44
- housing_units               1     285.6 14292 471.93
<none>                                     14007 472.16
- multi_unit_structure        1     619.2 14626 473.96
- single_parent_household     1     944.0 14951 475.90
- occupied_households         1    1045.5 15052 476.49
- total_population            1    1552.4 15559 479.41
```

```
- age_17_or_under              1    1999.1 16006 481.90
- no_hs_diploma                1    2307.4 16314 483.58
- below_poverty                1    2738.8 16746 485.87
- housing_type_transportation  1    2911.5 16918 486.78
- crowding                     1    3009.7 17016 487.29
- income                       1    4367.2 18374 494.04

Step:  AIC=471.44
`Primary Care Physicians Rate` ~ total_population + housing_units +
    occupied_households + below_poverty + income + no_hs_diploma +
    age_17_or_under + single_parent_household + housing_type_transportation +
    multi_unit_structure + crowding

                              Df Sum of Sq   RSS    AIC
<none>                                     14212 471.44
- housing_units                1     353.5 14566 471.60
- multi_unit_structure         1     679.0 14891 473.54
- single_parent_household      1     807.8 15020 474.30
- occupied_households          1    1035.2 15247 475.62
- total_population             1    1394.9 15607 477.68
- age_17_or_under              1    1848.1 16060 480.20
- no_hs_diploma                1    2233.3 16446 482.28
- below_poverty                1    2680.3 16892 484.64
- housing_type_transportation  1    3258.0 17470 487.60
- crowding                     1    3414.6 17627 488.39
- income                       1    5192.3 19404 496.84
```

Final Model

```r
summary(step_mod)
```

```

Call:
lm(formula = `Primary Care Physicians Rate` ~ total_population +
    housing_units + occupied_households + below_poverty + income +
    no_hs_diploma + age_17_or_under + single_parent_household +
    housing_type_transportation + multi_unit_structure + crowding,
    data = merged_data)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-24.207 -10.276  -1.199   9.152  34.792

Coefficients:
                             Estimate Std. Error t value Pr(>|t|)    
(Intercept)                -6.314e+01  1.743e+01  -3.622 0.000525 ***
total_population           -1.332e-03  4.877e-04  -2.731 0.007842 ** 
housing_units              -1.526e-03  1.110e-03  -1.375 0.173206    
occupied_households         3.714e-03  1.579e-03   2.353 0.021220 *  
below_poverty               3.423e-03  9.042e-04   3.786 0.000304 ***
income                      3.349e-03  6.355e-04   5.269 1.24e-06 ***
no_hs_diploma              -3.395e-03  9.823e-04  -3.456 0.000901 ***
age_17_or_under             3.523e-03  1.121e-03   3.144 0.002380 ** 
single_parent_household    -7.543e-03  3.629e-03  -2.078 0.041050 *  
housing_type_transportation  2.864e+01  6.861e+00   4.174 7.89e-05 ***
multi_unit_structure       -1.071e-03  5.623e-04  -1.906 0.060491 .  
crowding                   -2.821e-02  6.601e-03  -4.273 5.52e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.67 on 76 degrees of freedom
Multiple R-squared:  0.7546,	Adjusted R-squared:  0.7191 
F-statistic: 21.25 on 11 and 76 DF,  p-value: < 2.2e-16
```